ML Model Deployment Strategies and Model Serving using FastAPI

Objective

Train a Titanic classification model and demonstrate online, batch, streaming inference and REST API model serving.

1. Online vs Batch Inference

Online inference processes individual requests, whereas batch inference processes a group of records together.

2. Streaming vs Batch Inference

Streaming inference processes records as they arrive. Batch inference processes a collected group of records together.

3. Why FastAPI?

FastAPI exposes machine learning predictions through REST endpoints, supports Pydantic validation and provides interactive API documentation.

4. Why reuse the same model?

Reusing the same trained pipeline keeps preprocessing and predictions consistent across inference strategies and avoids unnecessary retraining.

Conclusion

The assignment demonstrates model training, evaluation, inference strategies and API-based model serving.